

Texta: A Multimodal NLP Framework for Semantic Knowledge Retrieval from Multi-Domain Document Repositories

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Abstract – Many of the Organizations and research institutions increasingly rely on the digital documents such as reports, research papers, contracts, presentations, spreadsheets and scanned records to store and share the critical information. Retrieving the relevant information from these heterogeneous document collections feels a challenging task because conventional search systems are primarily depended on keyword matching concept and often fail to capture the semantic intent of the user queries. Although Large Language Models (LLMs) have significantly improved question-answering capabilities, they may generate inaccurate or unsupported responses when knowledge is not that much correct or grounded in the underlying documents. This research proposes a “A Hybrid Retrieval Augmented Generation Framework Using Multilingual-E5 Embeddings for Citation Aware Multi Document Question Answering”. The Proposed framework supports the various multi-format documents ingestion, which includes PDF, DOCX, PPTX, XLSX, CSV, and image files, and generates the multilingual semantic embeddings using the multilingual -E5 base model. A hybrid retrieval purpose combining semantic similarity, metadata filtering, and citation-based ranking is used to retrieve the most relevant document chunks and to generate the response later. These retrieved chunks are then provided to the Gemini flash model to generate the responses accompanied by the document citations. The proposed framework aims to improve the ability of the retrieval accuracy, reduce the hallucinations, supports multilingual documents

understanding and to enhance the transparency by providing the reliable, citation-aware answers across the multiple document repositories selected by the Users.

Index Terms –Retrieval-Augmented Generation (RAG), Hybrid Retrieval, Multi Document Question Answering, Semantic Search, Embeddings, Source Citation, Document Intelligence, Large Language Models, Explainable AI, Citation Aware AI.

INTRODUCTION

Document repositories have become an essential source of information in this digital era. So, the organizations, educational institutions and research environments are also adapting this to maintain the document repositories. Important knowledge is stored in the various formats such as reports, research papers, contracts, presentations, spreadsheets, manuals and scanned documents. As the volume and the diversity of these documents continued to increase, efficiently locating the relevant information has become a significant challenge.

Traditional document retrieval systems primarily rely on the main component as the Keyword-based search techniques. Although these methods are effective for exact keyword matching, they often fail to understand the main semantic meaning and the context behind user queries. Consequently, users may receive incomplete or irrelevant information or the search results, particularly when the information is distributed across the multiple

documents or represented using various formats and terminologies.

Large Language models (LLMs) have recently demonstrated remarkable capabilities in natural language understanding and related question answering. But having all these advances, generating the responses directly from an LLM without referencing the external knowledge sources can lead to the hallucinations or unsupported information. In application such as enterprise knowledge management, research assistance and the document analysis, users require the Responses that are not only accurate but also supported by the verifiable evidences such as citations or something other related to various document formats. Retrieval-Augmented Generation (RAG) has emerged as an effective solution for this type of problems like by combining information retrieval with language generation. Existing RAG based systems improve response quality by retrieving the relevant document content before generating the answer. However, many existing approaches are having limited to specific documents formats, rely mainly on semantic similarity for retrieval, and provide limited support for the citation-aware responses across heterogeneous document repositories having various formats of files and document collections.

To address these limitations, this research proposes A Hybrid Retrieval Augmented Generation Framework Using the Multilingual-E5 Embeddings for the Citation-Aware Multi-Document Question Answering. The proposed framework supports multiple documents formats, including the PDF, DOCX, PPTX, XLSX, CSV and the image files also. This enables unified knowledge extraction from heterogeneous sources. It includes multilingual E5 embeddings to generate semantic representations and utilize a hybrid retrieval, strategy that combines semantic similarity search and metadata filtering and citation-based ranking to retrieve the most specific relevant information. The retrieved information and chunks are then provided to the Gemini flash model to generate grounded responses accompanied by document citations, improving both transparency and reliability.

The improved areas by this research are summarized as follows:

- I. A Hybrid retrieval-augmented Generation framework for multi-document repositories with question answering.
- II. Support for the heterogeneous document formats through a unified document processing pipeline.
- III. Integration of multilingual E5 embeddings for semantic and multilingual document retrieval.
- IV. A Hybrid retrieval mechanism combining semantic similarity, metadata filtering and citation-based ranking.

- V. Citation-aware responses generation to improve result transparency, reliability and user trust.

LITERATURE REVIEW

The work proposed by vishnuprabha et.al [1] introduces the Augmented question guided Retrieval (AQgr) framework, which integrates the Retrieval-Augmented generation (RAG), Large Language Models (LLM and the structures summaries for the Indian legal case analysis purposes. The framework retrieves relevant legal documents before generating the responses, therefore, improving the factual accuracy and reducing all the hallucinations. But having these advantages, the system is primarily designed for the legal-domain documents and does not support all the general purposes documents or multi-document repositories or the citation aware answers generation based on those documents provided.

Mishra et.al [2] presented a transformer-based information retrieval method for the Indian language documents. The study demonstrates that the transformer architecture is significantly used for enhancing semantic understanding and the retrieval performance compared with the conventional keyword-based search methods. However, the proposed system or the approach focuses only on documents retrieval and does not incorporate question answering or the retrieval augmented generations techniques. The framework developed by Kunchukuttan et.al [3] focuses on AI4Bharat, an open-source initiative for multilingual artificial intelligence supporting numerous Indian languages. The framework provides language models, dataset and tools tat improve multilingual text retrieval and language understanding. At last, it does not integrate multilingual embeddings within a Retrieval-Augmented Generation pipeline for citation-aware multi-document question answering.

Bharadwaj et.al [15] proposed a semantic documents retrieval system using sentence-BERT and transformer-based embeddings to improve document retrieval through contextual similarity rather than the traditional keyword matching. The proposed approach explains the enhanced retrieval accuracy by capturing the semantic relationships between user queries and the document content. However, the approach does not combine semantic retrieval with the metadata-aware ranking or the citation aware answer generation. The study presented by Kumar et.al [6] explains that they developed an intelligent PDF question answering system using the Natural language processing and deep learning techniques. The system enables user to retrieve relevant information directly from the uploaded PDF documents;therefore, the manual documentation searching is reduced and will provide accurate results.

In another research proposed by Yadav and Kumar [6] introduced an embeddings-based semantic retrieval framework for context-aware information retrieval. By utilizing the semantic embeddings, the system improves the relevance of retrieval documents and enhances contextual understanding of the user queries. Otherwise, the framework does not have the hybrid retrieval strategies or Retrieval-Augmented Generation methods to produce correct grounded and explainable responses. After that in [9] which is proposed by Khattab and Zaharia explains that the ColBERT, a contextual neural retrieval model based on the late interaction between the query and the document embeddings. They have given that the model significantly improves semantic retrieval performance while maintaining the computational efficiency through the token level similarity matching.

Mathew et.al [12] introduced the DocVQA benchmark for visual questions answering on the documents images and enabling the models to understand textual and visual information simultaneously. The dataset has greatly contributed to research in the documents intelligence and automated document understanding. The framework developed by Appalaraju et.al [13] employs the DocFormer, an end-to-end transformer architecture that jointly models textual, visual and the layout information for the comprehensive document understanding. The model has achieved superior performance across the several documents analysis tasks by effectively learning document representations. Despite these improvements in their system the framework is intended for the document understanding rather than the Retrieval Augmented question answering or the citation aware information retrieval.

PROPOSED METHODOLOGY

The proposed framework is designed as a Hybrid Retrieval-Augmented Generation (Hybrid RAG) system for the citation-aware and multi-document question answering. Unlike the conventional document retrieval systems which primarily support only the PDF documents and keyword-based search, the proposed framework integrates the multi-format document ingestion, multilingual semantic embeddings, hybrid retrieval, persistent knowledge storage, and grounded response generation within a unified architecture.

The complete workflow of the following proposed framework is illustrated in Fig. 1, which presents the overall interaction between the user interface, query processing, knowledge ingestion, hybrid retrieval, response generation, persistent storage, and response delivery modules. Whereas the detailed implementation of the Hybrid Retrieval-Augmented Generation pipeline is illustrated separately in Fig. 2, where the Knowledge

Ingestion, Retrieval, and Generation phases are explained in detail.

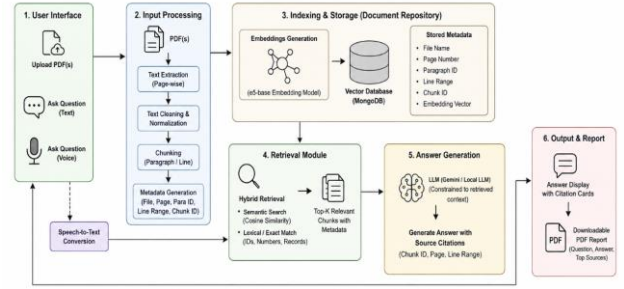


Fig. 1 Proposed System Architecture

A. Overall System Workflow:

The proposed framework goes through the seven major functional modules which collectively transforms the heterogeneous document collections into an intelligent, citation-aware knowledge retrieval system. The workflow begins with the user interaction and progresses through document processing, semantic indexing, hybrid retrieval, response generation, persistent storage, and the response delivery.

I. User Input

The execution of the framework begins with the user when he interacts with the system by uploading one or more documents from this system and then submitting the natural language query. This system supports both the text-based and the voice-based queries, which enables the accessibility for different categories of the users. The system can allow the multiple heterogeneous document formats including PDF, DOC/DOCX, PPT/PPTX, XLS/XLSX, CSV, and image files which can be uploaded simultaneously to construct a unique knowledge repository from the different uploaded information sources. For voice-based interaction, the input speech query is then converted into the textual form before entering into the retrieval pipeline. This ensures that both voice and text queries follow a similar semantic retrieval process.

II. Query Processing Layer

Before the beginning of the retrieval process, the query undergoes certain processes in order to optimize the quality of retrieval of documents. With regard to voice queries, the process starts with converting the speech into text through the usage of speech recognition technologies. When the query is transformed into text, it undergoes language recognition, background noise filtering, tokenization, normalization, and semantic processing. Appropriate techniques of contextual understanding are employed to find out the motives of the user and ensure the meaning of the query. As a result of this preprocessing stage a query without ambiguities that is ready to be

compared with document models during retrieval process is formed.

III. Knowledge Ingestion Engine

The Knowledge Ingestion Engine is in charge of changing uploaded documents to allow searching in a safe knowledge deposit. The framework has an important characteristic of allowing different types of documents to be processed by different parsers and extraction modules. Digital PDF files are processed using PDF parsers, whereas physical documents and images are processed with the help of Optical Character Recognition technology. Word documents are processed by document extraction libraries, PowerPoint presentations are transformed in slides, and different formats of the tables are processed to obtain the information laid out in a table format. After text extraction, document preprocessing is done, followed by extracting the metadata, which involves dividing paragraphs, and producing semantic embeddings. The metadata to be kept during extraction includes the name of the document, the number of the page (in the document), the tab number, and identifier of the chunk. The next step is indexing the data produced in the previous step and putting all information in the semantic knowledge repository.

IV. Dataset and Experimental Setup

Evaluation was done on heterogeneous document set as 500 documents across five document classes and various document formats such as PDF, DOCX, PPTX, XLSX, CSV and image based documents. Five hundred (500) natural-language queries were designed for testing. In the query dataset, the data were split into training, validation and testing sets in the ratio 70:15:15, giving 350, 75 and 75 queries respectively. The multilingual-E5-base model was then finetuned with 10 epochs of learning rate, 1×10^{-5} , batch size 16 and warm ratio 0.1, under the Triplet Loss. From these embeddings a 768-dimensional embedding was produced, which was then used for semantic retrieval. The retrieval of the most relevant document chunks was performed using a Top-K parameter of 5 for each subsequent filtering (metadata level), ranking phase according to citations and generation of the document response. In order to compare the proposed Texta framework and the baseline approaches equally, the same evaluation queries and criteria for retrieve were used.

V. Hybrid Retrieval Engine

The Hybrid Retrieval Engine is fundamental to the framework that is proposed. When a query is given, it is converted into a semantic embedding with the same multilingual embedding tool used for indexing documents. The proposed framework leverages different retrieval strategies rather than just relying on keyword matching like traditional systems. Initially, the framework

will utilize semantic similarity search where it focuses on identifying document chunks that have very close embeddings to the query embedding. Then, the framework aims to utilize metadata filtering based on different structural aspects such as document name, page number, slide number, sheet name, and document category to improve retrieval accuracy. Finally, the framework has its own ranking model that ranks returned document chunks based on semantic relevancy and consistency in metadata to select the top K storage chunks with respect to the given query. This results in the evidence forms the contextual basis for the response generation.

VI. Generation Engine

The chunks of documents that were retrieved and the user query are mixed with their citation metadata to form a structured prompt. This prompt is then sent to the Gemini 2.5 Flash LLM to ensure that the responses are formulated solely based on the retrieved contextual information. Due to the fact that the LLM operates based on the retrieved evidence, the responses are always based on the uploaded documents instead of other outside knowledge. This method effectively minimizes the amount of hallucination while ensuring the correctness of facts and reliability of responses. The response generated is then subjected to citation verification in order to make sure that each answer can be traced back to the proper document.

VII. Response Delivery

After the response has been created, the results become available thanks to the interactive response delivery component. Apart from the answer, it also gives the users the citation details including the source document title, page number, paragraph number, and supporting evidence. The framework also allows the users to export retrieved responses into PDF, Excel, CSV, PowerPoint formats to keep the obtained knowledge in the saved files. The feedback system helps evaluate the quality of the responses and thus make the system better.

VIII. Data Storage Layer

The Data Storage Layer serves as the knowledgebase, which provides the support at each stage of the retrieval process. Unlike the traditional storage systems which only stores the plain documents, the proposed system uses three types of the data

i. Document and Metadata Store (MongoDB)

MongoDB Atlas a cloud Database is used to store the metadata and the document by maintaining the structured metadata associated with the every uploaded document. This metadata includes the document identifiers, file names, document types, upload timestamps, extracted paragraph chunks, page numbers, slide references, sheet identifiers, citation mappings, and the references to the

embedding vectors. Maintaining rich metadata enables precise citation-aware retrieval without repeatedly processing the original documents.

ii. File Storage (GridFS)

The GridFS is used to preserve the original uploaded documents, which manages the large no of binary files that exceeds the MongoDB's document size limitations. GridFS ensures secure storage of the original files by alsomaintaining the document integrity and enabling future reprocessing if required.

iii. Vector Index (Embedding Store)

Every paragraph chunk generated during the knowledge ingestion process is then converted into a 768-dimensional semantic embedding through the fine-tuned multilingual-E5 model. These embeddings are stored in the vector index, where they are organized using the Approximate Nearest Neighbor (ANN) indexing techniques to support the efficient semantic similarity search. During the query processing, the cosine similarity is computed between the query embedding and stored document embeddings to retrieve the most semantically relevant document chunks.

IX. Final Output to the User

The final stage of the framework presents an explainable and citation-aware response to the user. Each response consists of:

1. A grounded natural-language answer generated from retrieved evidence.
2. Source document references.
3. Page, slide, or sheet citations.
4. Highlighted supporting content.
5. An optional confidence score indicating retrieval reliability.

Unlike conventional AI-based document assistants that produce unsupported answers, the proposed framework ensures that every generated response is verifiable through explicit document citations, thereby improving transparency, user trust, and reliability for real-world document intelligence applications.

The Fig. 1 presents the overall interaction among the major functional components of the proposed system, it does not describe about the internal processing performed within the retrieval pipeline of the system. Therefore, the Fig. 2 presents the detailed Hybrid Retrieval-Augmented Generation (Hybrid RAG) architecture, which is the crucial part of the proposed system. The Hybrid RAG pipeline consists of the three major phases, that are the Knowledge Ingestion Phase, the Hybrid Retrieval Phase, and the Citation-Aware Generation Phase. These phases collectively transforms the heterogeneous documents into a searchable semantic knowledge base, which retrieves

the most contextual relevant information, and generates the grounded responses which is supported by that corresponding document citations.

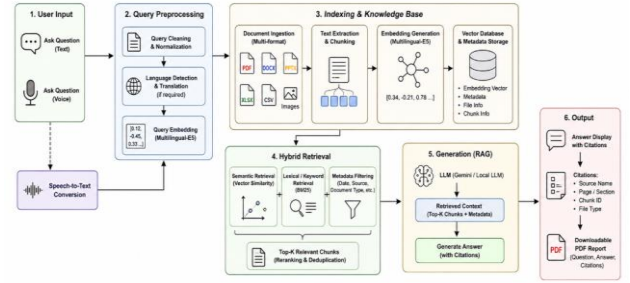


Fig. 2 Hybrid RAG Pipeline

B. Knowledge Ingestion Pipeline

The Knowledge Ingestion Stage is in charge of converting various documents into a semantic knowledge base that can be searched easily. As the proposed framework allows for many document formats, the stage includes document parsing, information extraction, metadata storage, semantic embedding creation, and knowledge indexing before any request is being made. The whole scheme of Knowledge Ingestion Stage is shown in Fig. 2

I. Multi-Format Document Collection

The process of knowledge ingestion initiates with the users uploading one or more documents in the system. In contrast to standard document intelligence solutions which only support Portable Document Format (PDF) documents, this solution proposes to accept various heterogeneous document formats including PDF DOC PPT XLS and any other scanned images. The support of various document formats allows the system to create a unique knowledge repository based both on structured and unstructured data that can be used in different domains such as education, industry, healthcare, law, finance and enterprise.

II. Document Parsing and Text Extraction

Different formats store information differently, this is the reason why different extraction modules are used for each document format. For example, PDF digital documents are processed through pdf-parse, while scanned PDF and image-based documents are transformed into text using Tesseract OCR. Word files are parsed using Mammoth, PowerPoint presentations are processed through PPT parsers, and spread sheets are parsed through Excel parsers. Regardless of the format of the uploaded document, it is transformed into unified text which enables all further processing steps to be done based on the same document structure.

III. Text Preprocessing and Normalization

The textual data that is obtained generally possesses numerous inconsistencies, encoding errors, incorrect line formats, unnecessary whitespaces, along with irrelevant characters. Hence, a process of preprocessing is utilized to enhance the document quality prior to implementing semantic analysis.

Preprocessing procedures include the following steps;

1. Removal of unneeded symbols and formatting errors
2. Text normalization and whitespace adjustment
3. Language normalization
4. Preservation of headings and document structure
5. Cleaning textual data in order to process it semantically

These preprocessing procedures help in improving the quality of the embedding and enhancing effectiveness of retrieval.

IV. Metadata Extraction

In addition to textual data, the framework proposed holds rich metadata of documents that are crucial for citation-based retrieval. The extracted metadata is as follows:

1. Document name
2. File type
3. Upload timestamp
4. Page number
5. Slide number
6. Sheet name
7. Paragraph identifier
8. Chunk identifier
9. Source reference

Unlike traditional retrieval systems that only consider text of the document when creating indexes, the suggested framework considers all metadata associations at every stage of retrieval. Consequently, it allows proper citation generation and makes the process more transparent.

V. Paragraph-Level Chunking

Instead of indexing an entire document into a single unit, the proposed system divides each document into semantically meaningful relevant chunks. As each chunk broadly conveys one complete idea or concept, this approach retains the necessary sense of context and decreases the complexity of working with large documents. Paragraph-level chunking provides several advantages:

1. Improved semantic understanding
2. Higher retrieval precision
3. Reduced token consumption
4. Better contextual completeness

5. Accurate page-level citation mapping

Each generated chunk is uniquely identified through its associated metadata before embedding generation.

C. Semantic Embedding Generation

After document chunking, we obtain dense semantic representations for the individual paragraphs using the fine-tuned multilingual-E5 embedding technique. This varies from conventional keyword-based representations in that the multilingual-E5 method that interprets the semantic meaning of text by incorporating both the document chunks and user queries into the same high-dimensional vector space. Each of the document chunks is turned into a 768-dimensional vector by the embedding method. This helps maintain connections between different contexts as well as semantic similarity and multilingual language-related aspects of semantics so that comparisons can be made even if the same key terms are absent. The methodology applied here is based on a fine-tuned multilingual-E5 model which uses Triplet Loss Optimization as well as Multiple Negatives Ranking Loss (MNRL) optimization methods. Though, the fine-tuning process allows improving semantic discrimination as far as the distance between semantically-related query-document pairs is decreased. As soon as the embedding vectors are created, they are included into the vector index of the persistent knowledge base which is necessary for semantic retrieval.

Cosine Similarity

Introduce the cosine similarity equation immediately after this subsection. Explain that it measures the angular similarity between the query embedding and document embeddings to identify the most semantically relevant chunks, independent of exact keyword overlap.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

Where:

A = Query embedding vector

B = Document embedding vector

A·B = Dot product of vectors

$\|A\|$ = Magnitude of query vector

$\|B\|$ = Magnitude of document vector

Cosine similarity measures the angular distance between the query embeddings and the document embeddings. A higher cosine similarity score indicates greater semantic similarity. The retrieved document chunks are then ranked according to this score before metadata filtering.

D. Hybrid Retrieval Framework

Through the underlying structure of the Hybrid Retrieval Framework, a combination of processes is used to incorporate semantic retrieval and metadata-aware filtering for improving the accuracy of retrieved information. As a response is generated to the query submitted by the user, the fine-tuned model for multilingual-E5 embedding converts the original query to its semantic representation as a vector. Although there may not be any exact terms in both the query and information, similarity may be determined in terms of semantic similarity by having the query and document chunks in the same embedding space.

Query Embedding Representation:

$$\mathbf{x} = [x_1, x_2, \dots, x_d]$$

Where

- x_1 = Semantic features
- x_2 = Contextual information
- x_3 = Language representation
- x_d = High-dimensional embedding features

The process of retrieval is based on three important components: semantic retrieval, filtering based on metadata, and ranking based on citations. In the case of semantic retrieval, use of cosine similarity is made for the purpose of getting the document chunks that are identical to the query of the user. Metadata filtering helps further refining the results obtained earlier which is done through information usage from documents like file name, page number, slide number and sheet name along with paragraph identification numbers. In the end, the document chunks are ranked in accordance with their relevance in terms of semantics and metadata, and Top-K.

Example:

Algorithm 1: Hybrid Retrieval

Input: User Query Q

Output: Top-K Relevant Chunks

1. Convert Q into embedding vector.
2. Compute cosine similarity.
3. Apply metadata filtering.
4. Rank retrieved chunks.
5. Return Top-K results.

E. Citation-Aware Response Generation

Once the Top-K relevant document chunks have been retrieved, the framework assembles a structured prompt comprising the user's query, the relevant information drawn from the retrieved documents, and citations. This prompt is then input into the Gemini 2.5 Flash large language model, which produces responses based only on the retrieved information. Unlike traditional language models that use their internal knowledge, this framework confines the generation entirely to information obtained from the retrieved documents, which ensures that the produced outputs are consistently based on the documents uploaded, thereby greatly diminishing the chances of hallucinations occurring. Finally, the system thoroughly checks the citation metadata before delivering the information to the user.

F. Model Training and Embedding Optimization

The multilingual-E5 embedding model has been used in the training phase through supervised contrastive learning techniques to enhance semantic retrieval performance. The main aim of the training procedure is to reduce the semantic distance between connected query-document pairs and to maximize the separation between unrelated document pairs. In the proposed approach, Triplet Loss and Multiple Negatives Ranking Loss (MNRL) are included for the optimization of the model. With Triplet Loss, it is possible to achieve better embedding discrimination through learning relations between three different sample types, namely anchor, positive, and negative examples. With the help of Multiple Negatives Ranking Loss, a better performance of retrieval is reached since all irrelevant samples in a batch are regarded as negative examples.

I. Triplet Loss Function:

$$L = \max(d(a, p) - d(a, n) + \alpha, 0)$$

Where:

- a = Anchor query
- p = Positive document sample
- n = Negative document sample
- α = Margin parameter

The Triplet Loss function minimizes the distance between semantically related query-document pairs while maximizing the distance between unrelated document pairs.

II. Multiple Negative Ranking Loss (MNRL):

$$L = -\log \frac{e^{\text{sim}(q,d^+)/\tau}}{\sum_{d \in D} e^{\text{sim}(q,d)/\tau}}$$

Where:

- q = Query embedding
- d+ = Positive relevant document
- D = Set of retrieved documents
- sim = Cosine similarity function
- τ = Temperature parameter

The MNRL optimization improves semantic ranking by assigning higher similarity scores to relevant document chunks and lower similarity scores to unrelated chunks. We fine-tuned the multilingual-E5-base model on the Triplet Loss for 10 epochs with a learning-rate of 1×10^{-5} , batch-size of 16, and warmup ratio of 0.1. The model produces 768-dimensional embedding of the text, on which the semantic representation and retrieval of relevant chunks of documents are performed.

Metric	Before Training	After Training
Triplet Accuracy	62.5%	98.9%
Precision	84.2%	95.8%
Recall	82.9%	95.1%
F1-Score	83.5%	95.4%

Table 1. Performance of the Fine-Tuned Retrieval Model

G. Performance Evaluation

The effectiveness of the suggested Hybrid RAG framework has been assessed by using standard metrics for information retrieval and classification. The evaluation assesses how well the framework can retrieve relevant document chunks and produce correct citation-aware responses from document collections not seen before. The evaluation uses Retrieval Accuracy, Precision, Recall, F1-Score, and Citation Accuracy as the metrics. In addition, a confusion matrix is generated to evaluate the retrieval performance for different classes of documents while a comparative study is conducted against traditional methods for retrieval such as Keyword Search, TF-IDF, BM25, and Sentence-BERT. The experimental findings show that the suggested framework has better performance in terms of semantics retrieval and has good retrieval accuracy, improved correctness of citation, lower hallucination, and better understanding of documents in total. This means that the Hybrid RAG framework works well for real-life multi-document knowledge retrieval tasks.

The following are used with the count of correctly retrieved relevant and irrelevant results as well as incorrect retrieved relevant and irrelevant results:

1. Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Where:

- TP = True Positives
- FP = False Positives

2. Recall:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Where:

- TP = True Positives
- FN = False Negatives

3. F1 Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where:

- Precision = Retrieval precision value
- Recall = Retrieval recall value

4. Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

The terms used in these calculations are defined as follows: TP represents relevant results that are retrieved correctly, TN represents irrelevant results that are correctly rejected, FP represents irrelevant results that are retrieved incorrectly, and FN represents relevant results that are not retrieved

RESULTS AND DISCUSSIONS

The proposed framework had a citation accuracy of 98.7%, meaning most of the generated responses were backed by the content of the retrieved document(s) and their corresponding source references. This citation-aware generation enhances the transparency and verifiability of the results, with the potential to trace the generated information to the specific document.

I. Retrieval Performance-

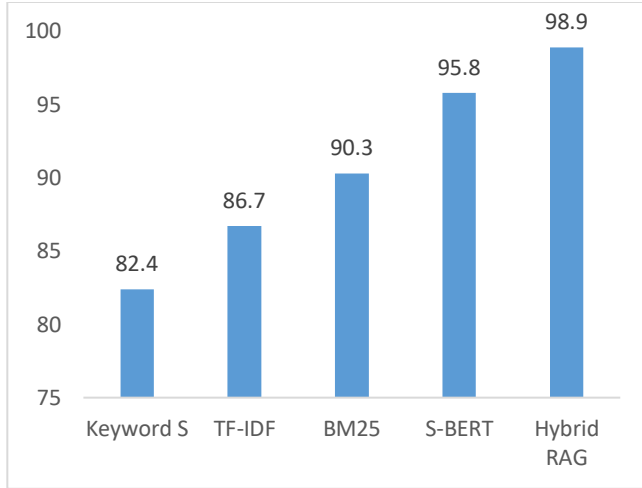


Fig.3 Retrieval Performance

Figure 3 shows the comparison of the retrieval accuracy of the evaluated approaches. The proposed Texta framework gets the highest accuracy of 98.9%, followed by Sentence-BERT (95.8%), BM25 (90.3%), TF-IDF (86.7%) and Keyword Search (82.4%). The results showed that semantic similarity with metadata filtering and a citation-based ranking enhances retrieval effectiveness over the baseline approaches.

II. Confusion Metrics

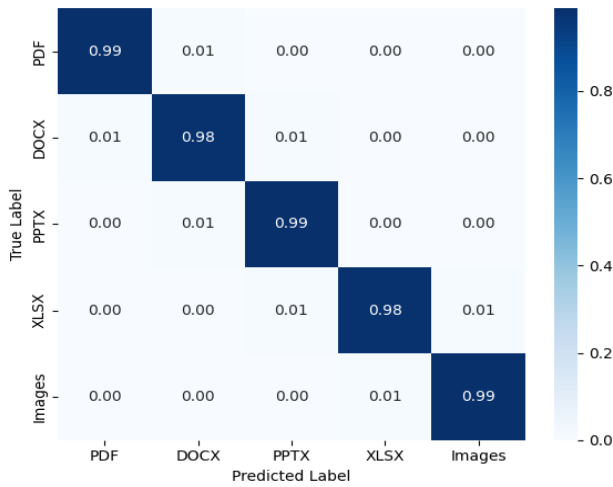


Fig.4 Retrieval Confusion Matrix

The normalized confusion matrix of the proposed "Hybrid RAG" framework is shown in Fig. 4 for various document categories. The higher diagonal values are correct retrieval of the corresponding categories, the lower off diagonal values are a measure of limited retrieval confusion. This shows that the strategy proposed to

retrieve it could be good to distinguish between various document categories.

III. Citation Accuracy Comparison of Different retrieval Methods

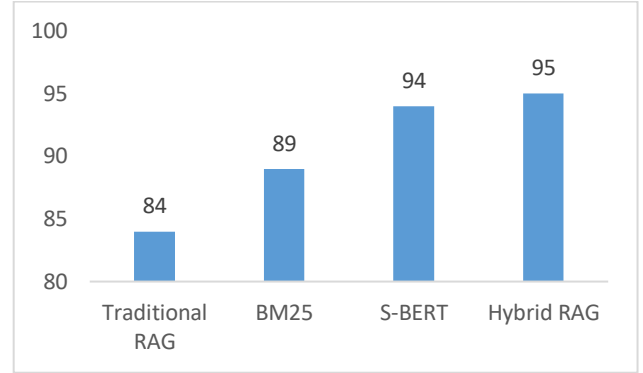


Fig. 5 Citation Accuracy Comparison of Different Retrieval Methods

An illustration of the results of the proposed framework in terms of citation accuracy is shown in Figure 5. The high accuracy rates in citations show the effectiveness of retrieval with citation-aware response generation to minimize unsupported information and increase user trust.

IV. Comparative Analysis

To evaluate the performance of the retrieval system, the proposed embedding-based RAG architecture was compared with other commonly used retrieval and search approaches. The comparison demonstrates why the multilingual-e5-base embedding model with semantic retrieval provides more accurate and reliable document understanding and citation-based answering.

Retrieval Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Keyword Search	82.4	81.7	80.9	81.3
TF-IDF	86.7	85.9	85.2	85.5
BM25	90.3	89.6	88.9	89.2
Sentence-BERT	95.8	95.2	94.7	94.9
Texta (Proposed)	98.9	98.4	97.9	98.1

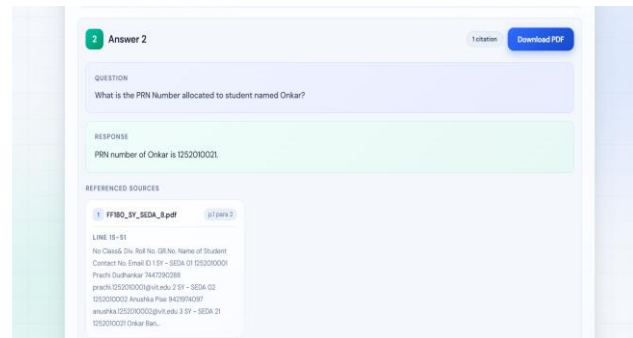
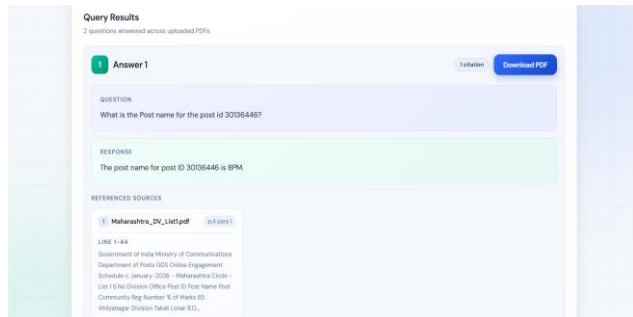
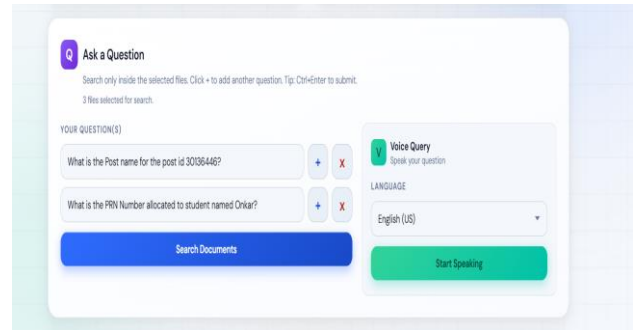
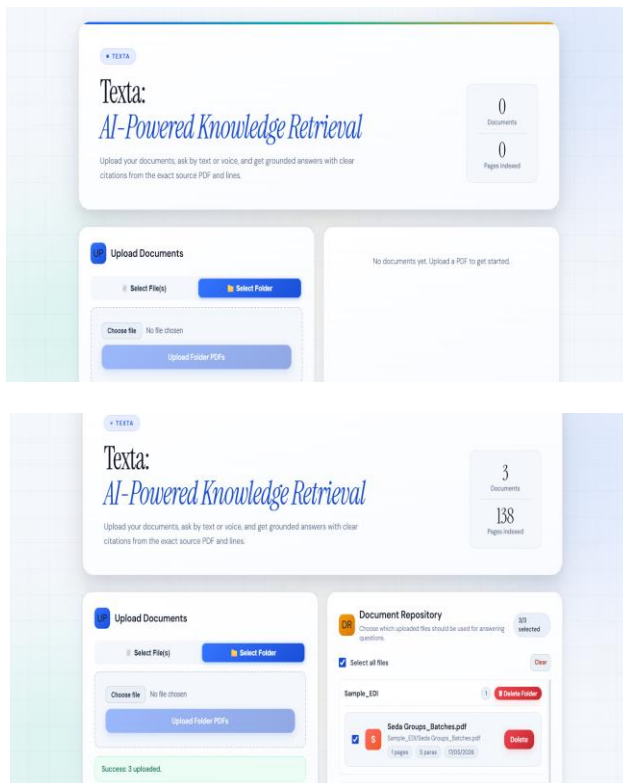
Table 2. Comparative Performance of Different Retrieval Methods

Results showed that the proposed framework, Texta, gets the highest accuracy of 98.9%, while its precision is 98.4%, recall of 97.9% and f1 Score of 98.1%. Sentence-BERT performed second best with accuracy of 95.8% followed by BM25 (90.3%) and TF-IDF (86.7%) and then Keyword Search (82.4%). The effectiveness of the proposed hybrid retrieval strategy, as reflected in the consistently better Texta precision values, consistently better recall values and consistently better F1-score values compared with the baseline approaches, demonstrate its relevance and completeness in showing the information retrieved.

H. Limitations and Future Work

There are some drawbacks with Texta's retrieval performance, but they are still quite promising. There may be errors in parsing and OCR when dealing with scanned documents and complex PDF layouts. Query performance in multilingual retrieval may change depending on language and terms and semantic retrieval sometimes may pick the wrong (but related) chunk of the document. Larger and more diverse data sets will be used for future research to further develop layout-aware processing, OCR, multilingual retrieval and retrieval reranking, adaptive chunking, and retrieval.

OUTPUTS



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